



Hospital Engineering and Management

Lecture 4: Nursing Unit and Intensive Care Unit Planning

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Learning Objectives

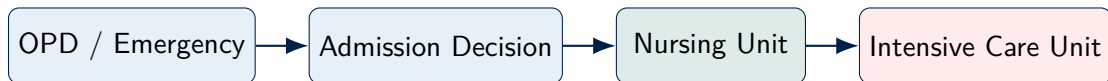
By the end of this lecture, students should be able to:

- ▶ explain the purpose and organization of a hospital nursing unit;
- ▶ identify the functional spaces and circulation requirements of nursing units;
- ▶ relate nursing workflow to patient-room and nursing-station design;
- ▶ describe the engineering services required in inpatient areas;
- ▶ distinguish general nursing units from intensive care units;
- ▶ explain the planning, equipment, safety and reliability requirements of an ICU.

Connection with the Previous Lecture

Lecture 3 Established

OPD planning focuses on efficient patient entry, consultation, diagnostics, support services and safe movement through a high-volume outpatient system.



Focus of Lecture 4

We now study the hospital areas where admitted patients receive continuous nursing care.

Opening Question: What Makes an Inpatient Area Safe?

Think About It

A patient may remain in a hospital bed for several days. What must function continuously around that patient, even when the doctor is not present?

- ▶ nursing observation;
- ▶ medicines and documentation;
- ▶ oxygen and suction when needed;
- ▶ electrical power and lighting;
- ▶ sanitation and infection control;
- ▶ food, linen and supplies;
- ▶ alarms and emergency response;
- ▶ safe movement of patients and staff.

The nursing unit is the patient's temporary living and care environment.

What Is a Nursing Unit?

Definition

A nursing unit is an organized inpatient area containing patient beds, nursing work areas, support rooms and utilities required for continuous patient care.

- ▶ It is the basic operational unit of inpatient hospital care.
- ▶ It connects patients with nursing, medical, diagnostic and support services.
- ▶ Its design should reduce unnecessary staff movement and improve observation.
- ▶ It must support privacy, comfort, infection control and emergency response.

Nursing Unit = Patient Space + Nursing Work + Support Services + Safe Circulation

Functions and Types of Nursing Units

Major Functions

- ▶ continuous patient observation;
- ▶ medication and treatment;
- ▶ documentation and communication;
- ▶ hygiene and basic nursing care;
- ▶ coordination with diagnostics and doctors;
- ▶ discharge preparation.

Common Types

- ▶ medical and surgical wards;
- ▶ pediatric and maternity units;
- ▶ orthopedic or specialty units;
- ▶ step-down or high-dependency units;
- ▶ isolation units;
- ▶ intensive care units.

Basic Components of a Nursing Unit

- ▶ patient rooms or bed bays;
- ▶ nursing station and work area;
- ▶ medication-preparation area;
- ▶ clean utility room;
- ▶ soiled utility room;
- ▶ equipment storage;
- ▶ linen and supply storage;
- ▶ toilets and bathing facilities;
- ▶ staff room and changing area;
- ▶ doctor / consultation workspace;
- ▶ visitor waiting space;
- ▶ housekeeping and waste holding area.

Planning Principle

Frequently used spaces should be located close to the point of care.

Functional Zoning of a Nursing Unit



Design Goal

Reduce staff travel while keeping public, clinical, clean and soiled activities appropriately separated.

Patient Room and Bed-Space Planning

A patient bed area should support both care delivery and patient comfort.

- ▶ adequate space for staff access around the bed;
- ▶ clear route for stretcher and wheelchair movement;
- ▶ access to electrical outlets, nurse call and medical gases as required;
- ▶ space for bedside monitor, infusion devices and mobile equipment;
- ▶ privacy for examination and personal care;
- ▶ hand-hygiene access and easy-to-clean surfaces;
- ▶ safe access to toilet and patient belongings;
- ▶ visual connection with staff without unnecessary loss of privacy.

Engineering Question

Can emergency equipment reach the patient quickly without moving furniture or other beds?

Single-Bed and Multi-Bed Rooms

Single-Bed Room

- ▶ better privacy;
- ▶ easier isolation;
- ▶ reduced cross-traffic;
- ▶ better family communication;
- ▶ higher space and infrastructure requirement.

Multi-Bed Room

- ▶ efficient use of floor area;
- ▶ easier observation of several patients;
- ▶ greater noise and privacy concerns;
- ▶ higher risk of flow conflicts;
- ▶ requires careful bed spacing and cleaning.

Room selection should reflect clinical need, infection risk, privacy and available resources.

Nursing Station Planning

The nursing station is a coordination point rather than merely a desk.

- ▶ supports observation, communication and documentation;
- ▶ connects nurses with doctors, laboratory, pharmacy and support staff;
- ▶ houses computers, communication devices and alarm interfaces;
- ▶ should not obstruct patient or trolley circulation;
- ▶ needs controlled access to clinical information and medicines;
- ▶ may be centralized, decentralized or hybrid depending on the unit design.

Design Trade-off

Centralized stations improve team communication; decentralized work points can reduce walking distance.

Visibility, Travel Distance and Response Time



- ▶ Poor layout increases staff walking and delays response to alarms.
- ▶ Good visibility improves supervision, but privacy must be preserved.
- ▶ Equipment and supplies should be located according to frequency of use.

Clean and Soiled Flow Separation

Clean Flow

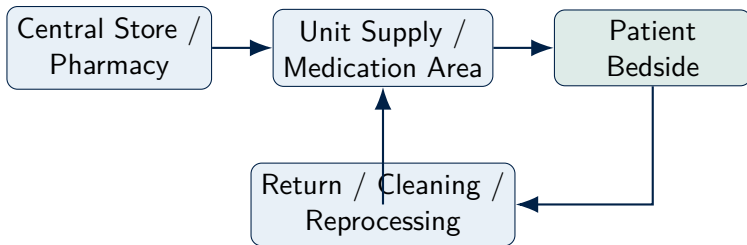
- ▶ sterile and clean supplies;
- ▶ medicines;
- ▶ clean linen;
- ▶ food and drinking water;
- ▶ disinfected equipment.

Soiled Flow

- ▶ used linen;
- ▶ clinical waste;
- ▶ contaminated equipment;
- ▶ specimens where applicable;
- ▶ cleaning materials.

Crossing clean and contaminated flows increases infection risk.

Medication, Supplies and Equipment Flow



Management Objective

Ensure the right item reaches the right patient at the right time while maintaining security, traceability and infection-control practices.

Nursing Services: Organization within the Unit

Nursing-unit design must support the actual work sequence of nurses.

- ▶ receive shift handover and review patient status;
- ▶ assess patients and record vital signs;
- ▶ prepare and administer medication;
- ▶ coordinate investigations and procedures;
- ▶ respond to nurse calls and clinical deterioration;
- ▶ maintain documentation and communicate with relatives;
- ▶ support discharge, transfer or escalation to intensive care.

Engineering Implication

A technically good layout is unsuccessful if it forces nurses to spend excessive time walking, searching for supplies or handling avoidable alarms.

Electrical and Communication Requirements

Electrical Services

- ▶ normal and essential power circuits;
- ▶ adequate bedside outlets;
- ▶ emergency lighting;
- ▶ safe earthing and protection;
- ▶ backup for critical devices;
- ▶ controlled use of extension boards.

Communication Services

- ▶ nurse-call system;
- ▶ telephone / intercom;
- ▶ hospital information system;
- ▶ clinical alarm communication;
- ▶ Wi-Fi / network connectivity;
- ▶ emergency-code notification.

Medical Gas and Suction Requirements

Medical-gas requirements depend on the clinical level of the nursing unit.

- ▶ oxygen may be required at selected or all bed locations depending on unit function;
- ▶ medical air and vacuum are more important in high-acuity areas;
- ▶ outlets must be clearly identified and protected against misuse;
- ▶ pipeline pressure and alarms require routine monitoring;
- ▶ reserve supply should support continuity during central-system interruption;
- ▶ hoses, flowmeters and suction regulators must be maintained and inspected.

Safety Principle

Medical gases are life-support utilities. Their availability, identification and alarm systems are safety-critical.

HVAC and Indoor Environmental Quality

HVAC design influences both comfort and infection control.

- ▶ maintain suitable thermal comfort for patients and staff;
- ▶ provide adequate ventilation and filtration;
- ▶ control humidity and unpleasant odours;
- ▶ avoid air movement that spreads contamination from high-risk areas;
- ▶ maintain special pressure relationships where isolation is required;
- ▶ provide access for filter replacement and preventive maintenance.

Important

Exact ventilation rates, pressure requirements and filtration levels must follow the applicable hospital design and infection-control standards.

Infection Prevention in Nursing Units

- ▶ accessible hand-hygiene points;
- ▶ cleanable surfaces and furniture;
- ▶ appropriate isolation capability;
- ▶ separation of clean and soiled materials;
- ▶ safe sharps and biomedical-waste disposal;
- ▶ routine cleaning schedules;
- ▶ equipment decontamination;
- ▶ controlled visitor movement;
- ▶ appropriate ventilation;
- ▶ staff adherence to standard precautions.

Design and Behaviour Work Together

Good infrastructure reduces infection risk only when supported by correct clinical practices.

Lighting, Acoustics and Ergonomics

Lighting

- ▶ examination light;
- ▶ comfortable ambient light;
- ▶ night lighting;
- ▶ emergency lighting.

Acoustics

- ▶ reduce alarm noise;
- ▶ limit corridor noise;
- ▶ support patient sleep;
- ▶ maintain communication.

Ergonomics

- ▶ safe patient handling;
- ▶ suitable work heights;
- ▶ reduce staff fatigue;
- ▶ clear equipment access.

Safety and Emergency Readiness

A nursing unit should be prepared for both individual patient emergencies and facility emergencies.

- ▶ rapid access to resuscitation equipment;
- ▶ unobstructed evacuation routes;
- ▶ fire detection, alarm and compartmentation provisions;
- ▶ emergency power and lighting;
- ▶ safe storage of chemicals and oxygen cylinders where used;
- ▶ fall-prevention measures and safe toilet design;
- ▶ clear emergency communication and escalation procedures.

Engineering Perspective

Safety should be built into the environment rather than added only after an incident occurs.

Biomedical Equipment in a Nursing Unit

- ▶ multiparameter monitors;
- ▶ infusion and syringe pumps;
- ▶ ECG machine;
- ▶ pulse oximeters;
- ▶ nebulizers;
- ▶ suction equipment;
- ▶ defibrillator / crash cart;
- ▶ portable oxygen equipment;
- ▶ glucometers;
- ▶ patient warmers where needed;
- ▶ mobility and transfer devices;
- ▶ specialty-specific equipment.

Biomedical Engineer's Role

Selection, installation, preventive maintenance, calibration, electrical-safety testing, user training and breakdown response.

What Is an Intensive Care Unit?

Definition

An intensive care unit is a specialized inpatient area for critically ill patients who require continuous observation, advanced monitoring and rapid life-support intervention.

- ▶ higher clinical acuity than a general ward;
- ▶ continuous or near-continuous monitoring;
- ▶ greater dependence on medical equipment and utilities;
- ▶ rapid access to trained staff and emergency treatment;
- ▶ stricter infection-control and reliability requirements.

ICU = Critical Care + Continuous Monitoring + Life Support + High Reliability

General Nursing Unit versus ICU

Feature	General Nursing Unit	ICU
Patient condition	Stable / moderate acuity	Critical / unstable
Observation	Periodic nursing assessment	Continuous monitoring
Equipment density	Moderate	High
Medical gases	According to unit need	Extensive bedside availability
Power continuity	Essential for selected loads	Critical for most bedside systems
Staff presence	Routine ward staffing	Higher staff intensity
Alarm load	Limited	High and time-critical
Space around bed	Routine care and transfer	Multiple staff and life-support devices

An ICU is an engineered high-dependency clinical environment.

Functional Zones of an ICU

Clinical Zone

- ▶ patient bays / rooms;
- ▶ nursing observation;
- ▶ medication and procedure areas;
- ▶ isolation rooms;
- ▶ emergency equipment.

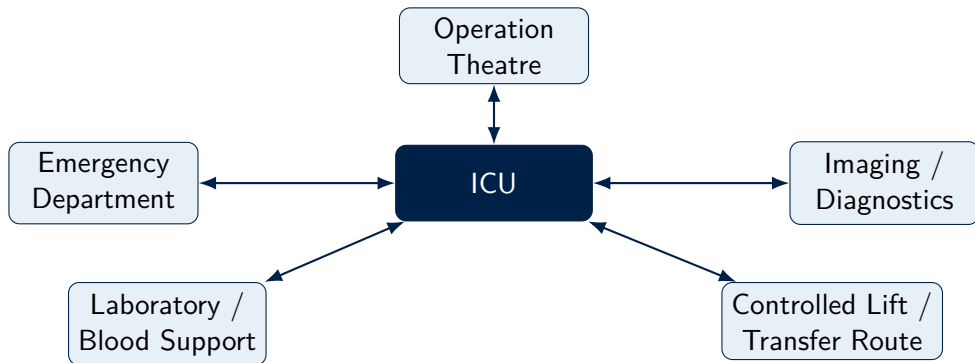
Support Zone

- ▶ clean and soiled utility;
- ▶ equipment store;
- ▶ staff work and rest areas;
- ▶ family communication / waiting;
- ▶ housekeeping and waste support.

Access Control

ICU circulation should limit unnecessary traffic while allowing rapid clinical and service access.

ICU Adjacency and Location



Planning Principle

Critical patients should reach essential diagnostic and treatment areas quickly with minimal transfer risk.

ICU Patient-Bay Planning

An ICU bed space must support simultaneous work by several clinicians and multiple devices.

- ▶ clear access to the patient's head and both sides of the bed;
- ▶ organized placement of ventilator, monitors and infusion systems;
- ▶ adequate electrical, communication and medical-gas connections;
- ▶ alarm visibility and audibility;
- ▶ space for portable imaging and emergency procedures;
- ▶ easy cleaning without inaccessible clutter;
- ▶ privacy while maintaining observation and rapid response;
- ▶ safe cable, tubing and hose management to reduce trip and disconnection risk.

ICU Monitoring, Alarms and Data Integration

Bedside Systems

- ▶ multiparameter monitor;
- ▶ ventilator;
- ▶ infusion / syringe pumps;
- ▶ defibrillator access;
- ▶ suction and oxygen;
- ▶ specialty life-support devices.

Information Flow

- ▶ central monitoring;
- ▶ clinical documentation;
- ▶ laboratory results;
- ▶ alarm communication;
- ▶ equipment status;
- ▶ hospital information systems.

Human-Factors Concern

Too many non-actionable alarms can distract staff and reduce the effectiveness of genuinely critical alarms.

Reliability and Redundancy in ICU Engineering

ICU services must be designed to tolerate foreseeable failures.

- ▶ essential and backup electrical supply for life-support equipment;
- ▶ UPS support where uninterrupted transfer is required;
- ▶ reliable medical-gas supply with alarms and reserve arrangements;
- ▶ redundant or standby critical pumps and infrastructure where appropriate;
- ▶ continuous environmental monitoring for critical HVAC conditions;
- ▶ backup communication and emergency-response procedures;
- ▶ preventive maintenance scheduled to avoid interruption of patient care.

In critical care, a single infrastructure failure must not become a patient catastrophe.

Isolation and Infection Control in ICU

Critically ill patients are particularly vulnerable to healthcare-associated infection.

- ▶ isolate patients according to infection-control requirements;
- ▶ maintain appropriate air-pressure relationships for designated isolation rooms;
- ▶ provide hand hygiene close to each point of care;
- ▶ minimize unnecessary equipment sharing;
- ▶ clean high-touch surfaces and reusable devices systematically;
- ▶ manage invasive lines, ventilator circuits and suction equipment safely;
- ▶ control staff and visitor movement during outbreaks.

Important

ICU infection control combines architectural design, engineering controls, equipment practices and clinical behaviour.

Case Study: Converting a Ward into a Temporary ICU

During a sudden increase in critically ill patients, a hospital proposes converting part of a general medical ward into a temporary intensive-care area.

Discussion Questions

1. Is the existing electrical supply adequate for multiple life-support devices?
2. Are oxygen and suction available at every required bed?
3. Can the HVAC system support the intended infection-control strategy?
4. Is there sufficient space for staff, equipment and emergency access?
5. How will alarms, monitoring and backup power be managed?

Engineering Analysis of the Case

Potential Risks

- ▶ overloaded electrical circuits;
- ▶ inadequate oxygen capacity;
- ▶ poor ventilation / isolation;
- ▶ insufficient bedside space;
- ▶ alarm and network limitations;
- ▶ delayed equipment maintenance.

Required Actions

- ▶ engineering load assessment;
- ▶ medical-gas capacity verification;
- ▶ HVAC and infection-control review;
- ▶ emergency-power testing;
- ▶ equipment and alarm planning;
- ▶ multidisciplinary commissioning.

Classroom Activity

Activity: Design a Safe Nursing Unit

Work in small groups. Sketch a simplified nursing unit for 20 admitted patients.

Your diagram should show:

1. patient rooms or bed bays;
2. nursing station;
3. clean and soiled utility areas;
4. medication / supply area;
5. staff and visitor support;
6. emergency equipment location;
7. one isolation provision;
8. one route for transfer to ICU or diagnostics.

Mini Project Ideas

- ▶ nursing-unit workflow and staff-walking-distance audit;
- ▶ bedside electrical-outlet and equipment-load assessment;
- ▶ nurse-call system reliability study;
- ▶ ICU alarm-management improvement proposal;
- ▶ preventive-maintenance plan for ward or ICU equipment;
- ▶ clean / soiled circulation audit of an inpatient unit;
- ▶ ICU utility-readiness checklist for emergency expansion;
- ▶ patient-room safety and accessibility audit.

Expected Output

Identify a real operational problem, collect basic observations and propose a practical engineering or management intervention.

Summary

- ▶ A nursing unit integrates patient beds, nursing work, support spaces and utilities.
- ▶ Layout should reduce travel distance while supporting observation, privacy and infection control.
- ▶ Electrical power, communication, medical gases, HVAC and sanitation are essential engineering services.
- ▶ Biomedical engineers support equipment availability, safety, calibration and integration.
- ▶ ICU patients require continuous monitoring, life-support capability and higher infrastructure reliability.
- ▶ ICU planning must consider location, bed-space organization, alarms, redundancy and infection control.

Good inpatient design brings care, equipment and support to the patient safely and quickly.

Assignment for the Next Lecture

Study any hospital nursing unit or ICU through observation, a published layout, institutional information or a virtual hospital tour.

Prepare a one-page note containing:

1. a simple functional layout;
2. location of the nursing station;
3. clean and soiled utility flow;
4. five important biomedical equipment items;
5. three essential engineering utilities;
6. one patient-safety or infection-control observation;
7. one practical improvement suggestion.

Submission Format

One page, individually prepared, with a labelled diagram.

Suggested Reading

- ▶ G. D. Kunders, *Hospitals: Facilities Planning and Management*, Tata McGraw-Hill.
- ▶ B. M. Sakharkar, *Principles of Hospital Administration and Planning*, Jaypee Brothers Medical Publishers.
- ▶ S. Goel et al., *Textbook of Hospital Administration: A Problem-Solving Approach*, Elsevier India.

Next Lecture: Nursing Services – Organization, Staffing and Quality of Care

Thank You

Questions and Discussion